Decision-Time Provenance for Multimodal Fact-Checking: Independent Evidence, Temporal Validity, and Calibrated Deferral

Sabbir Hossain
Department of Computer Science
American International University-Bangladesh
Dhaka, Bangladesh
26-94030-1@student.aiub.edu

***Abstract*—Genuine images can support false claims when they are paired with the wrong event, place, or date. Web retrieval can recover the missing context, but common evaluation pipelines admit evidence that was unavailable when a decision was made and count syndicated copies as independent corroboration. They can also force a verdict when the available record is inadequate. This study introduces DTP-Fact, a framework for classifying image–text claims as supported, refuted, or insufficient evidence. Its primary contribution is a decision-time evidence protocol: an item enters the main analysis only when both its publication metadata and an independent archive or crawl observation precede the claim-specific cutoff. The framework groups copied and linked documents into source-dependence clusters, bounds each cluster's aggregate contribution, and records the evidence path in a typed provenance graph. A hierarchical predictor estimates evidence sufficiency before veracity; temperature scaling and a validation-selected threshold provide selective deferral. The accompanying US-TPMFC design uses United States ClaimReview cases from 2020–2026, with chronological, event-disjoint splits and separate reporting for the partial 2026 snapshot. Counterfactual tests inject syndicated copies and post-cutoff verdicts, remove early sources, and vary dependent and independent conflicts. The protocol compares matched content-only, context-recovery, relevance-reranking, document-level RAG, and cluster-aware systems. No empirical superiority is claimed before the preregistered 200-case feasibility audit and frozen real-data experiments are complete.**

***Index Terms*—multimodal misinformation, automated fact-checking, evidence provenance, temporal retrieval, source dependence, selective prediction**

I. Introduction

Digital misinformation often reuses genuine media outside its original context. A photograph from one event may be paired with a claim about another place, date, or incident. Pixels alone may show no manipulation, and image–text compatibility can identify a mismatch without establishing what the image originally depicted. NewsCLIPpings and VERITE demonstrate both the importance of false-context detection and the danger of unimodal shortcuts [1], [2]. A fact-checking system therefore needs external evidence that connects the claim and image to documented events.

Retrieval supplies that context, but a valid historical decision depends on when the evidence became observable and where it originated. For a claim assessed at cutoff time τ, a later correction, official report, or fact-check page contains information that was unavailable at the decision point. A self-reported publication field is not enough to prove availability: a page first captured after τ may have been backdated, changed, or unavailable to the contemporary system. Search results can also give a false impression of corroboration. Syndicated articles, reposts, and quotation chains may all trace to one report, so document-level aggregation can overweight one underlying source.

Evidence availability is different from model confidence. The task uses three verdict labels. *Supported* means that time-valid evidence corroborates the claim, *refuted* means that it contradicts the claim, and *insufficient evidence* means that the available material cannot settle the claim. Selective abstention is a separate model decision: the system may defer any prediction when its calibrated confidence is too low. This separation permits independent measurement of evidential sufficiency and decision reliability.

Recent systems address parts of this setting. MOCHEG joins multimodal retrieval and verification [3]; RED-DOT reranks web evidence by relevance [4]; and COVE reconstructs an image's original context before assessing its caption [5]. ChronoFact models temporal relations within claims and evidence [6], while work on conflicting evidence shows that hard credibility filtering can remove useful counter-evidence [7]. None of these components alone establishes that retrieved evidence was available at the claim-specific decision time or that repeated documents represent independent sources.

This study asks whether decision-time-valid evidence from independent source clusters improves future-event fact-checking and selective risk relative to matched document-level multimodal RAG. Each case is an English claim about a United States event, paired with an image and a decision cutoff. DTP-Fact admits only evidence with a documented pre-cutoff observation, groups dependent documents, represents the resulting trail in a typed graph, and estimates evidence sufficiency before veracity. Source

dependence is not treated as source credibility: the primary model limits duplicate influence without suppressing evidence because of a publisher-level trust score.

The study makes three contributions:

- a decision-time construction protocol for US-TPMFC that records timestamp confidence, excludes unobserved pre-cutoff evidence from the primary analysis, and isolates events, near-duplicate claims, and reused images by split;
- a source-dependence-aware framework that bounds each evidence cluster's aggregate mass, separates audited sufficiency supervision from inherited verdicts, and applies calibrated selective deferral; and
- a falsifiable evaluation using future events and counterfactual evidence interventions for post-cutoff leakage, duplicate inflation, early-source removal, and dependent versus independent conflicts.

II. Related Work

A. Multimodal Misinformation Detection

NewsCLIPpings introduced a large benchmark of authentic news images paired with deliberately mismatched captions [1]. VERITE later addressed unimodal bias and topic imbalance in real-world evaluation [2]. KAMP uses single-modality, cross-modal, and knowledge-content objectives during multimodal pretraining [8]. Other work uses synthetic examples to improve transfer to real misinformation cases [9]. Event-Radar represents visual manipulation, textual emotion, and event-level inconsistency in an event graph [10]. These methods provide the main no-retrieval and event-aware baselines in this study.

Uncertainty-aware fusion is another relevant content-only line. MDF-FND represents modality uncertainty and changes the contribution of text and images dynamically [11]. Propagation-based methods instead use how a post spreads; RPPG-Fake generates early heterogeneous propagation paths [12]. Neither setting establishes the origin or historical availability of external web evidence, but both motivate controlled uncertainty and early-information comparisons.

B. Evidence-Grounded Fact-Checking

MOCHEG treats multimodal fact-checking as evidence retrieval followed by verification and explanation [3]. MetaSumPerceiver summarizes evidence from several multimodal documents [13]. RED-DOT addresses noisy web evidence through relevance-aware reranking [4]. RAGAR and MultiReflect combine multimodal language models with iterative retrieval [14], [15], while SNIFFER joins internal image–text consistency with external image context [16]. These studies motivate strong document-level retrieval and reasoning baselines. They do not test whether a large result set reflects several independent sources or copies of one source.

C. Image Context, Time, and Evidence Dependence

The 5Pils system recovers an image's original context from web evidence and excludes evidence that follows the fact-check [17]; COVE predicts that context before assessing a caption [5]. Chronological distribution shift can substantially reduce multimodal misinformation performance [18]. ChronoFact handles temporal relations within complex textual claims through timeline reasoning [6]. These works concern context, event time, or future generalization. The present protocol focuses on a different temporal variable: the earliest independently observed availability of each evidence item relative to a claim-specific cutoff.

Source information also affects decisions under conflicting evidence [7]. Because source filtering can discard useful disagreement, DTP-Fact separates dependence from credibility. Exact and near-duplicate content, shared imagery, direct links, and publication order support dependency clustering; publisher reputation does not determine primary-model inclusion. The joint object of study is therefore decision-time-valid independent corroboration, rather than a new graph encoder or a general source-trust score.

III. Dataset and Task

A. Task Definition

Each instance contains claim text x_t, an image x_v, the earliest observed claim time, a decision cutoff τ, and an evidence collection $E_{\leq\tau}$. The output is $y \in \{\text{supported}, \text{refuted}, \text{insufficient evidence}\}$. Selective deferral is an operational decision applied after calibration, not a fourth class. The event is the independent unit for splitting and statistical analysis.

B. Candidate Acquisition

Corpus construction starts with English ClaimReview records about United States events from 2020–2026. GDELT supplies discovery metadata, and Common Crawl supplies archived page observations. The claim time is the earliest documented public appearance. The decision cutoff is the fact-check publication time or a recorded equivalent. The corpus retains cases with a recoverable image or archive, an unambiguous professional rating, and enough timestamp information to enforce that cutoff. Fact-check pages, their domains, and verdict-bearing post-cutoff documents are excluded from the primary evidence pool.

The first authenticated candidate snapshot was frozen at 2026-10-24 07:53:33 UTC. It contains 32,898 deduplicated ClaimReview candidates and 440 preserved API response pages from seven configured publisher filters. This number is a screening-pool count rather than a benchmark size: eligibility, United States relevance, image identity, rating, event membership, and decision-time metadata still require audit. PolitiFact, AFP, and Snopes reached the preregistered 100-page collection cap, so this snapshot is not interpreted as a census of their ClaimReview records.

Evidence timestamps have three confidence levels. *Confirmed* items have consistent publication metadata and an

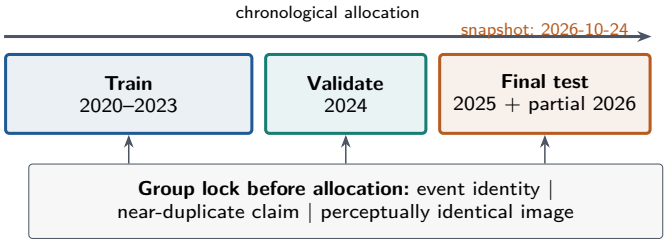


Fig. 1. Primary temporal split and leakage controls. A locked event, near-duplicate claim, or reused image cannot cross partitions.

independent pre-cutoff observation. *Observed* items have a pre-cutoff archive or crawl time but incomplete publication metadata. *Uncertain* items rely on self-reported metadata or were first observed after the cutoff. The primary analysis uses confirmed and observed items only; uncertain items enter a sensitivity analysis. A post-cutoff archive does not establish pre-cutoff availability.

C. Label Policy

Only explicit site ratings are normalized through a frozen, site-specific mapping. True or correct maps to supported; false, fake, or miscaptioned maps to refuted; and unproven, unsubstantiated, or no evidence maps to insufficient evidence. Mixed, partially true, satire, opinion, and ambiguous ratings are excluded. The labels come from professional fact-checkers, and automatically extracted evidence links remain silver evidence.

The evidence-sufficiency target is audited separately. It is not inferred solely from the professional verdict: a later supported or refuted verdict does not prove that enough time-valid evidence existed at an earlier cutoff. A 200-case pilot receives gold sufficiency judgments under a written protocol. Any model trained with automatically derived sufficiency targets reports those labels as silver and evaluates them separately.

D. Temporal Partitioning and Release

Training covers 2020–2023, validation uses 2024, and final testing uses 2025 plus a partial-2026 snapshot ending on 2026-10-24. The 2025 and 2026 results are reported separately and jointly. Events, near-duplicate claims, and images with matching perceptual hashes remain in one partition. The release package contains URLs, hashes, timestamp confidence, normalized labels, graph annotations, retrieval queries, and split manifests. It does not redistribute copyrighted articles or images.

Figure 1 shows the order of operations. Group identities are fixed before the year-based allocation. Training data are used to fit model parameters and preprocessing; validation data are reserved for model selection, calibration, and abstention thresholds; the final period is not used for any of those choices.

This study treats US-TPMFC as a benchmark only if the filtered corpus contains at least 2,000 cases, 300 cases per label, 50 test events, and two independent evidence clusters for at least 70% of retained claims. If the collection misses any threshold, it becomes a temporal and provenance stress test. The threshold is fixed before inspecting the full collection.

IV. Methodology

A. Framework Overview

Figure 2 follows one claim from retrieval to a traceable decision. The cutoff controls which evidence may enter the graph, while dependency clustering controls how that evidence is counted. Both constraints act before classification. Calibration cannot repair a prediction formed from future or duplicated evidence.

For claim c and evidence pool E, the retained set is

$$E_{\le\tau}^{(k)} = \mathrm{TopK}_k\{r(c,e) : e \in E,\ p(e) \le \tau,\ o(e) \le \tau\}, \quad (1)$$

where $r(c,e)$ is the fused retrieval score, $p(e)$ is the reported publication time, $o(e)$ is the earliest independent archive or crawl observation, and $k = 20$. The primary set excludes uncertain timestamps. The provenance stage maps the remaining items to dependency clusters $C(E_{\le\tau}^{(k)})$ before graph encoding.

B. Time-Constrained Multimodal Retrieval

For each claim, retrieval begins with named entities, temporal expressions, OCR text, and an image caption. BM25 and BGE-M3 retrieve text candidates, while SigLIP2 embeddings and perceptual hashes retrieve images. Reciprocal-rank fusion merges lexical, dense, and visual rankings. A cross-encoder reranks the top 100 candidates. Eligibility filtering then retains at most 20 items whose publication and observation times satisfy the cutoff. All model revisions, queries, candidate lists, and filtering reasons are preserved.

C. Evidence-Source Provenance Graph

The implemented graph contains nodes for claims, images, documents, publishers, and dependency clusters. Typed edges record claim–image membership, retrieved evidence, publication, cluster membership, image containment, direct links, and publication order; reverse relations are added for message passing. Clusters are estimated from exact text hashes, token-set title similarity, perceptual image distance, and direct hyperlinks. Publication order is retained as a graph relation rather than treated as proof that the earliest item is the original source.

Let $C(E)$ denote the partition of evidence documents into dependency clusters. If cluster j contains n_j documents, document i in that cluster receives base mass $w_i = 1/n_j$. A learned within-cluster aggregator may redistribute that mass, but the cluster's total contribution is bounded. The model then combines information across clusters. Ten copies of one report therefore cannot receive the same base weight as ten independently sourced reports. Cluster quality is evaluated on a manual pair sample before the main experiment.

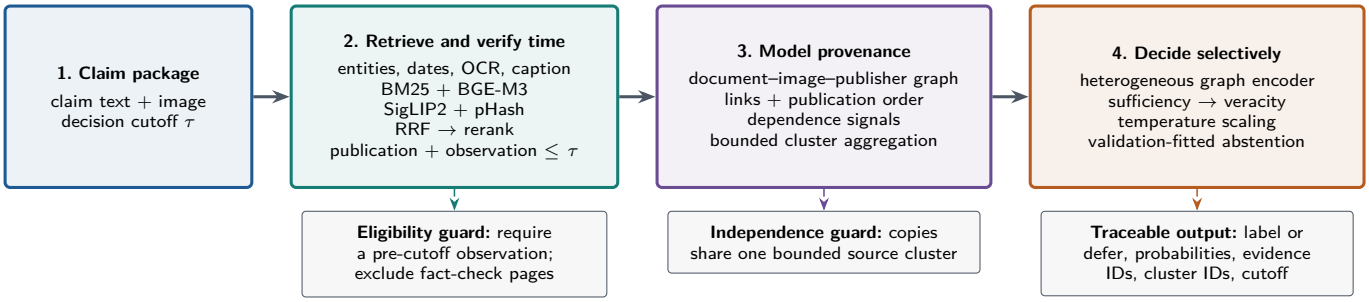


Fig. 2. End-to-end DTP-Fact pipeline. The temporal guard removes unavailable evidence, provenance clustering prevents copies from being treated as independent corroboration, and the final record preserves the evidence trail. Selective deferral is applied to the calibrated three-class prediction and is not a fourth task label.

Source dependence is kept separate from source credibility. Domain identity can define a publisher node but does not exclude or downweight an item in the primary model. This choice preserves contradictory evidence while preventing repeated publication chains from inflating corroboration.

D. Hierarchical Decision and Abstention

A relation-aware graph encoder represents the temporal-provenance graph. The sufficiency head estimates $p(s = 1 \mid x, E_{\leq\tau})$ using separately audited sufficiency targets. Cases predicted as sufficient pass to a veracity head for supported or refuted; the remainder receive the insufficient-evidence label. A flat three-class model and a parameter-matched cluster-pooling model test whether the hierarchy and graph are necessary.

Let z_s be the sufficiency logit and (z_+, z_-) the supported/refuted logits. The implemented three-class distribution is

$$p(y = \text{supported}) = \sigma(z_s)\ \text{softmax}(z_+, z_-)_+, \quad (2)$$

$$p(y = \text{refuted}) = \sigma(z_s)\ \text{softmax}(z_+, z_-)_-, \quad (3)$$

$$p(y = \text{insufficient}) = 1 - \sigma(z_s). \quad (4)$$

Training minimizes class-weighted negative log likelihood. When a separately audited sufficiency target is available, a masked binary cross-entropy term is added with validation-selected weight; unlabeled sufficiency rows do not contribute to that auxiliary term. Evidence-node feature dropout is applied during full-model training and is removed in a named ablation.

Temperature scaling is fitted to validation logits, and a threshold selected on the same split targets 80% answer coverage. Deferral can apply to any class when calibrated confidence is below the threshold. Each prediction records probabilities, selected evidence, dependency clusters, cutoff, timestamp policy, and whether the system answered or deferred.

V. Experimental Setup

A. Baselines

The executable controlled comparison includes majority prediction, text-only classification, image–text fusion, vanilla multimodal RAG, time-filtered RAG, provenance-aware cluster pooling, a parameter-matched no-graph model, and the full relation-aware model. These methods share the same frozen records, feature encoders, splits, evaluation code, and 20-trial validation budget. The controlled ablations remove one temporal, provenance, modality, hierarchy, graph, or regularization component from the same implementation.

MDF-FND, Event-Radar, an open multimodal language model, COVE, and RED-DOT are secondary comparators only where their public implementations and input assumptions can be reconciled with the frozen protocol [11], [10], [5], [4]. An incompatible or unavailable method is reported as such rather than approximated and ranked as a controlled reproduction. Literature-reported values remain separate because they use different data and splits.

B. Metrics and Statistical Analysis

The primary metric is macro-F1 on the event-disjoint future test set, reported for 2025, partial 2026, and both periods combined. Secondary metrics are per-class F1, balanced accuracy, macro one-versus-rest AUROC, Brier score, negative log-likelihood, expected calibration error, area under the risk–coverage curve, risk at 80% coverage, Recall@10, nDCG@10, latency, GPU memory, and retrieved-document count. Dependency clustering also reports pairwise precision, recall, and F1 on the audit sample. Final configurations use five seeds. Differences use 10,000 paired event-level bootstrap replicates with Holm correction across the primary comparison family.

C. Ablation and Stress Tests

The ablations remove observation-time filtering, dependency clustering, temporal edge features, the graph encoder, text evidence, visual evidence, the sufficiency gate, and evidence-dropout training one at a time. The graph ablation uses parameter-matched cluster pooling rather than raw document pooling. Timestamp sensitivity admits uncertain items without changing any trained parameter.

Figure 3 shows the two principal counterfactual tests. Duplicate injection adds 1, 5, or 10 linked copies of one report. Leakage injection adds a post-cutoff document containing the verdict. Further tests remove the earliest observed

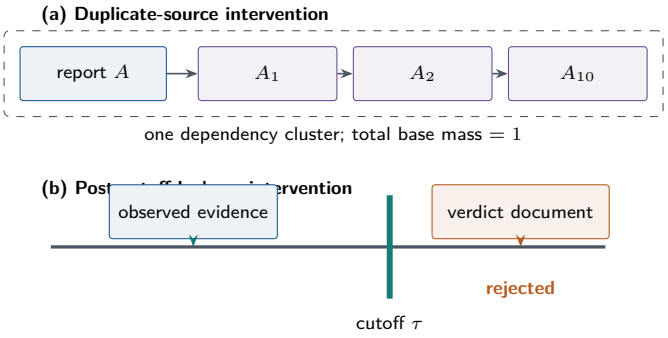


Fig. 3. Counterfactual evidence interventions. Duplicate injection should not increase independent corroboration, while the decision-time gate must reject a label-revealing post-cutoff document.

TABLE I
PREREGISTERED CLAIM-TO-TEST MATRIX.

Question	Controlled test and rejection condition
Decision-time validity	Full model vs. vanilla and provenance-only RAG; uncertain-date and post-cutoff interventions. Reject a performance claim if any advantage exists only with leaked evidence.
Copy robustness	Bounded cluster vs. document aggregation with 1, 5, and 10 copies. Reject if confidence, Brier score, selective risk, and macro-F1 degrade similarly.
Sufficiency first	Hierarchical vs. flat three-class prediction on audited cases. Reject if insufficient-evidence F1 does not improve or matched-coverage risk increases.
Calibrated deferral	Temperature-scaled vs. uncalibrated confidence. Reject if risk at 80% coverage or AURC does not improve at matched coverage.

source, introduce conflicts among dependent and independent clusters, swap captions, and reuse historical images. Conflict donors are selected deterministically from opposite-label cases, but they enter the reported analysis only after the pilot audit confirms that they are semantically opposed to the target claim; otherwise an audited donor manifest replaces the automatic pairing. Each intervention writes a manifest so the modified evidence set can be reconstructed exactly.

D. Selection and Falsification Rules

Hyperparameters and the deferral threshold are selected using validation data only. Decision-time provenance must improve valid future-event performance or duplicate robustness to support a performance claim. Calibrated deferral must reduce selective risk at matched coverage. Gains that appear only when post-cutoff documents are admitted count as leakage. If the candidate corpus misses any preregistered scale or coverage threshold, it is reported as a stress test rather than a benchmark.

Table I links each intended claim to a controlled comparison and an explicit rejection condition. This keeps the interpretation tied to tests that can fail rather than to a favorable result chosen after evaluation.

E. Implementation and Reproducibility

BGE-M3 encodes claim and evidence text, while SigLIP2 encodes verified local images. Exact public model revisions are fixed in the configuration before feature extraction. AdamW optimization uses class weighting, early stopping on validation macro-F1, and a cosine learning-rate schedule. Every selection trial writes its command, validation score, checkpoint, and log; final configurations are rerun with seeds 11, 23, 37, 53, and 71. The code records the Python, PyTorch, CUDA, GPU, precision, elapsed time, peak GPU memory, checkpoint hash, calibration temperature, and abstention threshold. Test predictions are exported only by the final-run command, whereas the tuning command does not construct test examples.

VI. Results

The real-data feasibility pilot and frozen model runs have not yet been completed. Consequently, this draft reports no accuracy, calibration, retrieval, or efficiency result, and it makes no superiority claim. Synthetic smoke-test outputs are software diagnostics and are excluded from the manuscript's scientific evidence.

The completed study will report three result blocks generated directly from preserved artifacts. The first covers corpus yield, exclusions, timestamp confidence, label balance, sufficiency-audit agreement, and dependency-cluster quality. The second covers retrieval, three-class prediction, calibration, and selective risk for the matched baselines. The third reports every counterfactual intervention, ablation, topic slice, and chronological window, including negative or inconclusive findings. The 2025 and partial-2026 results remain separate in addition to the combined test result.

VII. Discussion

A. Interpretation

The central hypothesis concerns the evidence set, not the complexity of the graph encoder. A positive result requires an advantage under valid future-event evidence or a measurable reduction in sensitivity to copied sources. If parameter-matched cluster pooling performs as well as the graph model, independent-source aggregation remains relevant but the graph contribution must be removed. If cutoff filtering helps only when the unrestricted baseline receives label-revealing future documents, the study supports a leakage warning rather than a general accuracy claim.

B. Ethical Considerations

The corpus inherits professional verdicts and may reproduce institutional or topical biases in the selected fact-checkers. The system is intended to assist review, not to assign reputational penalties or automate content removal. Public release is limited to code, URLs, hashes, timestamps, derived annotations, and split manifests unless source licenses permit redistribution. Search queries and archives may expose deleted or sensitive material; collection logs therefore require access controls and a documented takedown procedure.

C. Limitations

The planned corpus is English-language and United States-specific. A single-author 200-case audit checks extraction and protocol adherence but does not establish inter-annotator reliability. Archive coverage is incomplete, and first observed time is an upper bound on public availability rather than proof of original publication. Dependency clustering cannot always recover the true reporting chain, especially when copying is paraphrased or offline. The 2026 test is a partial-year snapshot. Live web retrieval is nonstationary, and model comparisons may differ in pretraining and tool access even under a shared candidate pool. These limits constrain geographic, temporal, and causal generalization.

VIII. Conclusion

This study defines a test of decision-time provenance in multimodal fact-checking. The proposed protocol requires independently observed pre-cutoff evidence, limits the influence of dependent copies, separates evidence sufficiency from selective deferral, and evaluates these choices on future events through reconstructable interventions. The contribution will stand only if the feasibility gate is met and the corresponding comparisons support it. Until those experiments are complete, DTP-Fact remains an implemented research protocol rather than a validated fact-checking system.

References

[1] G. Luo, T. Darrell, and A. Rohrbach, "NewsCLIPpings: Automatic generation of out-of-context multimodal media," in *Proceedings of EMNLP*, 2021, pp. 6801–6817.

[2] S.-I. Papadopoulos, C. Koutlis, S. Papadopoulos, and P. C. Petrantonakis, "VERITE: A robust benchmark for multimodal misinformation detection accounting for unimodal bias," *International Journal of Multimedia Information Retrieval*, vol. 13, p. 4, 2024.

[3] B. M. Yao, A. Shah, L. Sun, J.-H. Cho, and L. Huang, "End-to-end multimodal fact-checking and explanation generation: A challenging dataset and models," in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023, pp. 2733–2743.

[4] S.-I. Papadopoulos, C. Koutlis, S. Papadopoulos, and P. C. Petrantonakis, "RED-DOT: Multimodal fact-checking via relevant evidence detection," *IEEE Transactions on Computational Social Systems*, vol. 12, no. 6, pp. 4231–4240, 2025.

[5] J. Tonglet, G. Thiem, and I. Gurevych, "COVE: Context and veracity prediction for out-of-context images," in *Proceedings of NAACL-HLT*, 2025, pp. 2029–2049.

[6] A. M. Barik, W. Hsu, and M. L. Lee, "ChronoFact: Timeline-based temporal fact verification," in *Proceedings of IJCAI*, 2025, pp. 8031–8039.

[7] Z. Ge, Y. Wu, D. W. K. Chin, R. K.-W. Lee, and R. Cao, "Resolving conflicting evidence in automated fact-checking: A study on retrieval-augmented LLMs," in *Proceedings of IJCAI*, 2025, pp. 9656–9664.

[8] L. Zhang, X. Zhang, Z. Zhou, X. Zhang, P. S. Yu, and C. Li, "Knowledge-aware multimodal pre-training for fake news detection," *Information Fusion*, vol. 114, p. 102715, 2025.

[9] F. Zeng, W. Li, W. Gao, and Y. Pang, "Multimodal misinformation detection by learning from synthetic data with multimodal LLMs," in *Findings of EMNLP*, 2024, pp. 10 467–10 484.

[10] Z. Ma, M. Luo, H. Guo, Z. Zeng, Y. Hao, and X. Zhao, "Eventradar: Event-driven multi-view learning for multimodal fake news detection," in *Proceedings of ACL*, 2024, pp. 5809–5821.

[11] H. Lv, W. Yang, Y. Yin, F. Wei, J. Peng, and H. Geng, "MDF-FND: A dynamic fusion model for multimodal fake news detection," *Knowledge-Based Systems*, vol. 317, p. 113417, 2025.

[12] L. Zhang, X. Zhang, Z. Zhou, X. Zhang, S. Wang, P. S. Yu, and C. Li, "Early detection of multimodal fake news via reinforced propagation path generation," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 2, pp. 613–625, 2025.

[13] T.-C. Chen, C.-W. Tang, and C. Thomas, "MetaSumPerceiver: Multimodal multi-document evidence summarization for fact-checking," in *Proceedings of ACL*, 2024, pp. 8742–8757.

[14] M. A. Khaliq, P. Y.-C. Chang, M. Ma, B. Pflugfelder, and F. Miletić, "RAGAR, your falsehood radar: RAG-augmented reasoning for political fact-checking using multimodal large language models," in *Proceedings of the Seventh Fact Extraction and VERification Workshop*, 2024, pp. 280–296.

[15] U. Kangur, K. Agrawal, Y. Singh, A. Sabir, and R. Sharma, "MultiReflect: Multimodal self-reflective RAG-based automated fact-checking," in *Proceedings of MAGMaR*, 2025, pp. 1–17.

[16] P. Qi, Z. Yan, W. Hsu, and M. L. Lee, "SNIFFER: Multimodal large language model for explainable out-of-context misinformation detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 13 052–13 062.

[17] J. Tonglet, M.-F. Moens, and I. Gurevych, ""image, tell me your story!" predicting the original meta-context of visual misinformation," in *Proceedings of EMNLP*, 2024, pp. 7845–7864.

[18] N. Stepanova and B. Ross, "Temporal generalizability in multimodal misinformation detection," in *Proceedings of the 1st GenBench Workshop on (Benchmarking) Generalisation in NLP*, 2023, pp. 76–88.